

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

D O P 3 1 9

Zero-downtime deployment strategies

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Deployment strategy

In the next few years, organizations will build over **500**

A deployment strategy is a way to **change** or **upgrade** an application; the aim is to make the change without downtime in a way that the users **barely notice** the changes
million new apps, more than the number developed in the previous 40 years combined

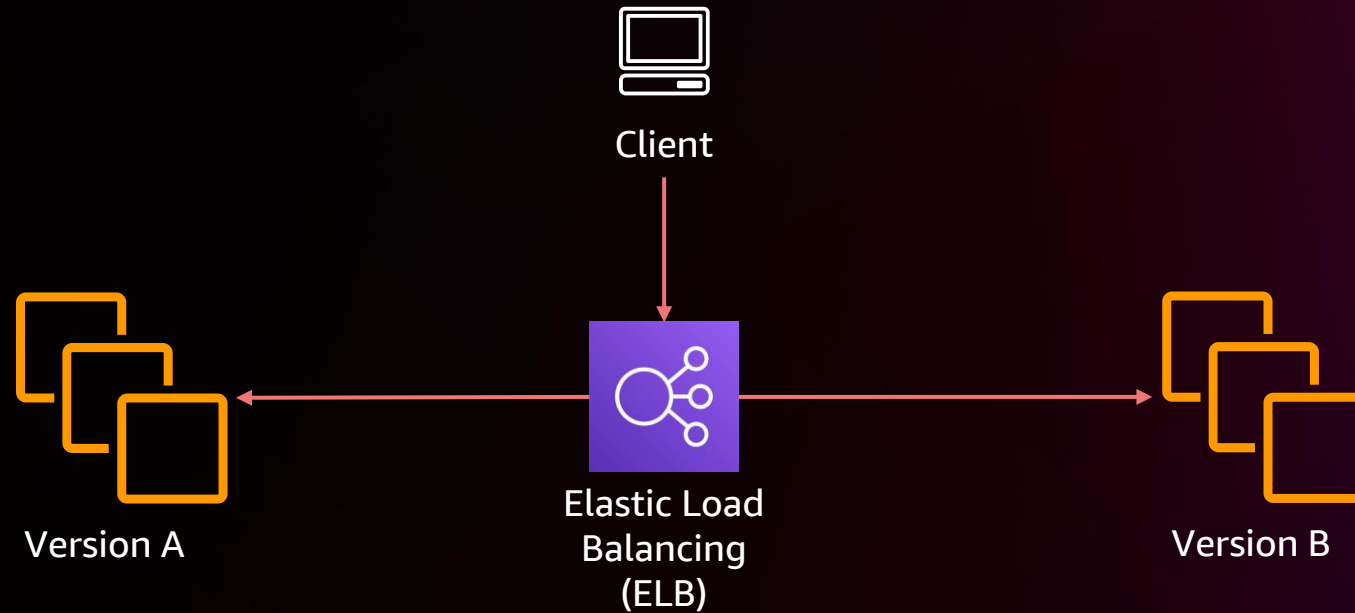
Source: IDC



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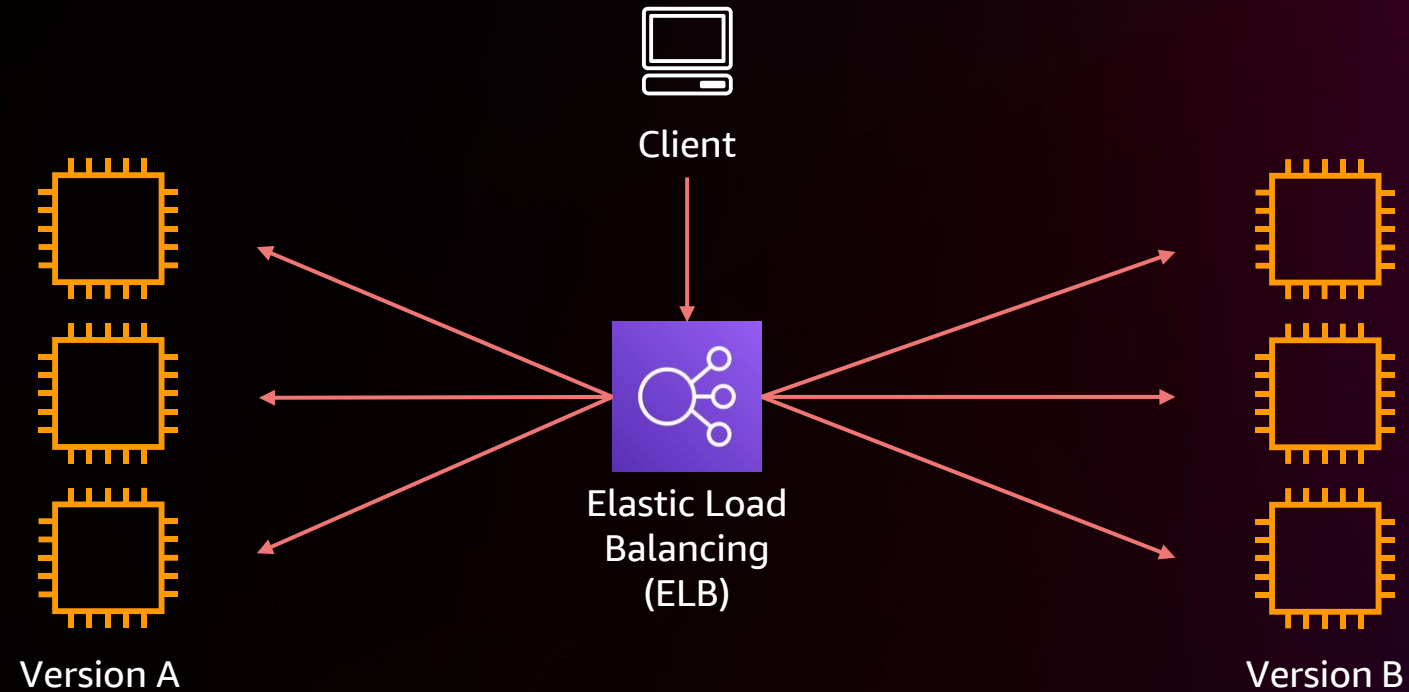
All at once

Version A is terminated, then version B is rolled out



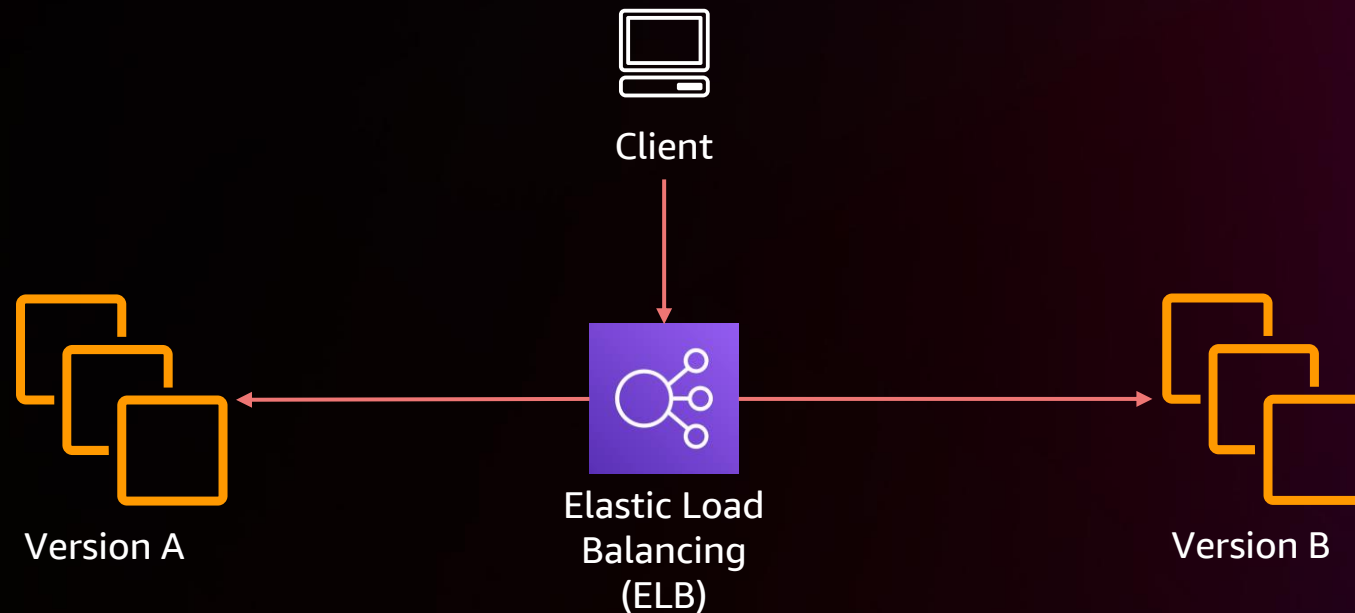
Rolling update

Version B is slowly rolled out and replaces version A



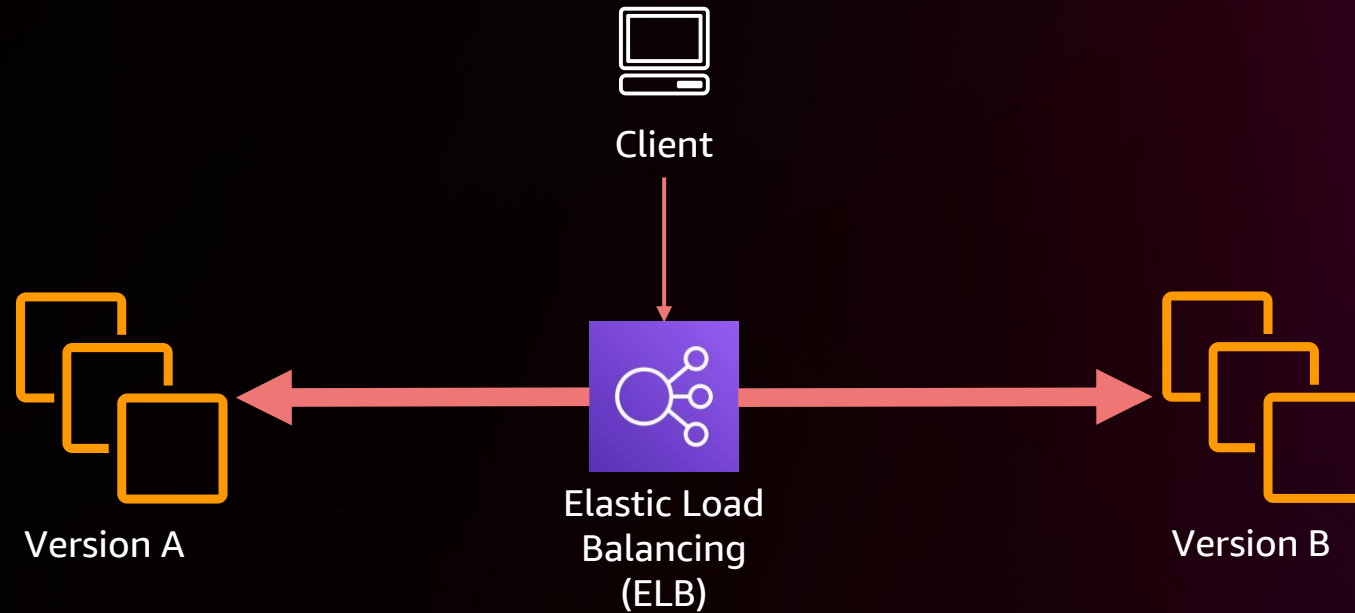
Blue/green

Version B is released alongside A, and then the traffic is switched to version B



Canary

Version B is released to a subset of users and then proceed to a full rollout



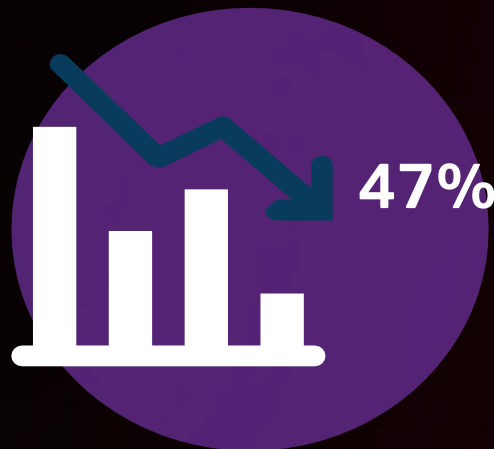
Why zero downtime

Unplanned application downtime costs the **Fortune 1000** from **\$1.25 billion** to **\$2.5 billion** every year

Source: IDC



Lost revenue



Lost productivity



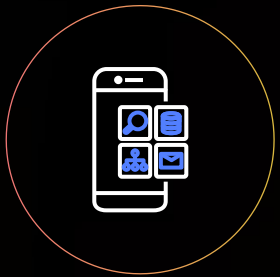
Lost brand equity or trust

Source: Forrester



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Architecting for zero downtime



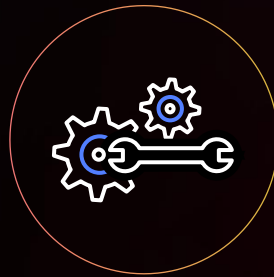
Monitor

Create monitors and notification channels across all tiers



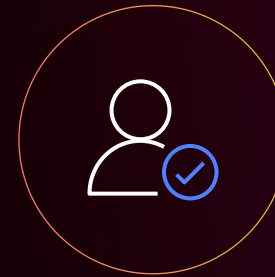
Resiliency

Ensure enough instances are always available



Configuration

Tune settings for planned recovery scenarios



Validate

Test successful and failure conditions repeatedly



Simplify

Automate and script as much as possible

Zero downtime with Amazon EKS



Thank you!

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